

KKT First-Order Conditions

Chapters 6 and 7 developed first- and second-order conditions for unconstrained optimization, while Chapters 8 to 11 developed numerical methods that seek stationary points. All of that theory presumes that every sufficiently small displacement is available. A constraint removes directions and can exert a reaction force, so a constrained minimizer may have a nonzero objective gradient. This chapter replaces unconstrained stationarity by the Karush–Kuhn–Tucker (KKT) first-order balance law. When this algebraic system is used as a necessary optimality test under a constraint qualification, we call it the *KKT first-order necessary conditions (FONC)*.

Learning objectives

After this chapter, you should be able to:

- explain why an unconstrained stationary-point condition or update can fail at a constrained optimum;
- interpret KKT first-order stationarity, multiplier signs, and complementary slackness as force balance and contact conditions;
- define a KKT point and write the Lagrangian, the complete KKT first-order system, and its active-set form;
- distinguish the tangent cone from the linearized cone and prove their basic relationship;
- state and verify LICQ, and explain its separate roles in multiplier existence and uniqueness;
- derive the KKT FONC from tangent geometry and a Farkas alternative;
- solve modest KKT first-order systems by active-set enumeration without confusing a KKT point with a certified minimizer.

12.1 From unconstrained stationarity to constrained optimization

12.1.1 A boundary optimum with nonzero gradient

We begin with an example used earlier in the course.

Example 12.1 (A closest point on a polygon). Consider

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & f(\mathbf{x}) := (x_1 - 4)^2 + (x_2 - 6)^2 \\ \text{subject to} \quad & h_1(\mathbf{x}) := x_1 - 4 \leq 0, \quad h_2(\mathbf{x}) := x_2 - 6 \leq 0, \\ & h_3(\mathbf{x}) := 3x_1 + 2x_2 - 18 \leq 0, \\ & h_4(\mathbf{x}) := -x_1 \leq 0, \quad h_5(\mathbf{x}) := -x_2 \leq 0. \end{aligned} \quad (12.1)$$

The unique global minimizer is

$$\mathbf{x}^* = \left(\frac{34}{13}, \frac{66}{13} \right)^\top.$$

To verify this claim without using the KKT first-order conditions, observe that

$$\nabla f(\mathbf{x}^*) = \left(-\frac{36}{13}, -\frac{24}{13} \right)^\top = -\frac{12}{13}(3, 2)^\top.$$

For every feasible $\mathbf{x}$, the third constraint and $3x_1^* + 2x_2^* = 18$ imply

$$\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) = -\frac{12}{13}(3x_1 + 2x_2 - 18) \geq 0.$$

Since f is a quadratic with Hessian $2I$,

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) + \|\mathbf{x} - \mathbf{x}^*\|^2 \geq f(\mathbf{x}^*),$$

with equality only at $\mathbf{x} = \mathbf{x}^*$. Hence $\mathbf{x}^*$ is indeed the unique global minimizer, even though $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$.

Moreover, an ordinary steepest-descent step is infeasible for every positive step length. Since $h_3(\mathbf{x}^*) = 0$,

$$\begin{aligned} h_3(\mathbf{x}^* - \alpha \nabla f(\mathbf{x}^*)) &= h_3(\mathbf{x}^*) - \alpha \nabla h_3(\mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) \\ &= 12\alpha > 0 \quad (\alpha > 0). \end{aligned}$$

Thus the negative gradient points through the active boundary.

Warning

The failure of a raw unconstrained step does not mean that Newton, steepest descent, or conjugate-gradient ideas are useless for constrained problems. It means that they cannot be applied unchanged. Later methods restore feasibility or encode the constraints through projections, penalties, barriers, or constrained subproblems.

12.1.2 The smooth nonlinear program

Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open, and suppose $f, g_i, h_j : \mathcal{D} \rightarrow \mathbb{R}$ are continuously differentiable. We study

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}), \quad \mathcal{S} := \{\mathbf{x} \in \mathcal{D} : g_i(\mathbf{x}) = 0 \ (i = 1, \dots, m), \ h_j(\mathbf{x}) \leq 0 \ (j = 1, \dots, p)\}. \quad (12.2)$$

The equalities describe two-sided surfaces. With the fixed convention $h_j \leq 0$, an inequality describes a one-sided wall and ∇h_j points toward increasing violation.

Definition 12.1 (Constrained local and global minimizers). A feasible point $\mathbf{x}^*$ is a local minimizer of Equation (12.2) if there is $\varepsilon > 0$ such that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for every } \mathbf{x} \in \mathcal{S} \cap B_\varepsilon(\mathbf{x}^*).$$

It is a strict local minimizer if the inequality is strict for every $\mathbf{x} \neq \mathbf{x}^*$ in that neighborhood. It is a global minimizer if

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for every } \mathbf{x} \in \mathcal{S}.$$

Definition 12.2 (Active set). For $\mathbf{x} \in \mathcal{S}$, the active inequality set is

$$J(\mathbf{x}) := \{j \in \{1, \dots, p\} : h_j(\mathbf{x}) = 0\}.$$

An index outside $J(\mathbf{x})$ is inactive.

Important note: Only feasible comparisons matter

The definition of a constrained local minimizer compares $\mathbf{x}^*$ only with nearby points of $\mathcal{S}$. A direction that decreases f is irrelevant if it immediately leaves $\mathcal{S}$. This is why $\nabla f(\mathbf{x}^*) = \mathbf{0}$ is no longer a necessary condition at a boundary point.

12.2 KKT first-order conditions as force balance

12.2.1 One physical model, three contact regimes

Let $\mathbf{x} = (x_1, x_2)$ and, for a parameter $c \in \mathbb{R}$, consider

$$f_c(\mathbf{x}) := \frac{1}{2}(x_1 - c)^2 - x_2, \quad g(\mathbf{x}) := x_2 = 0, \quad h(\mathbf{x}) := x_1 - 1 \leq 0. \quad (12.3)$$

The objective is the potential energy of a horizontal spring anchored at $x_1 = c$ together with a constant upward load. The equality is a frictionless horizontal rail, and the inequality is a rigid wall at $x_1 = 1$ that keeps the particle on its left.

The feasible set is the half-line

$$\mathcal{S} = \{(x_1, 0) : x_1 \leq 1\},$$

and restriction of the objective to $\mathcal{S}$ gives

$$f_c(x_1, 0) = \frac{1}{2}(x_1 - c)^2.$$

Therefore

$$\mathbf{x}_c^* = \begin{cases} (c, 0), & c \leq 1, \\ (1, 0), & c > 1. \end{cases} \quad (12.4)$$

The relevant gradients are

$$\nabla f_c(\mathbf{x}) = (x_1 - c, -1)^\top, \quad \nabla g(\mathbf{x}) = (0, 1)^\top, \quad \nabla h(\mathbf{x}) = (1, 0)^\top.$$

If λ and ν denote reaction coefficients, the physical forces are

$$\mathbf{F}_f = -\nabla f_c, \quad \mathbf{F}_g = -\lambda \nabla g, \quad \lambda \in \mathbb{R}, \quad \mathbf{F}_h = -\nu \nabla h, \quad \nu \geq 0. \quad (12.5)$$

At equilibrium their sum is zero. Solving

$$-\nabla f_c(\mathbf{x}_c^*) - \lambda^* \nabla g(\mathbf{x}_c^*) - \nu^* \nabla h(\mathbf{x}_c^*) = \mathbf{0}$$

gives

$$\lambda^* = 1, \quad \nu^* = c - x_{c,1}^* = \max\{c - 1, 0\}.$$

Consequently:

c	$\mathbf{x}_c^*$	wall status	ν^*
2	(1, 0)	active and loaded	1
0	(0, 0)	inactive	0
1	(1, 0)	active but unloaded	0

An illustration for the case $c = 2$ is in Figure 12.1.

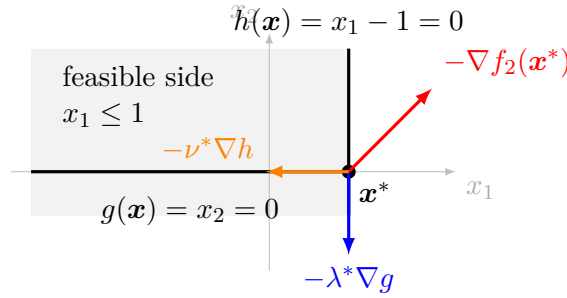


Figure 12.1: For $c = 2$, the objective, rail, and wall forces balance at the loaded contact point. Every physical force is written as a negative gradient.

Important note: Forces and KKT first-order signs

With $h_j \leq 0$, the outward normal is ∇h_j , while the physical wall reaction is $-\nu_j \nabla h_j$ with $\nu_j \geq 0$. The physical equilibrium equation

$$-\nabla f - \sum_i \lambda_i \nabla g_i - \sum_j \nu_j \nabla h_j = \mathbf{0}$$

becomes the standard KKT first-order stationarity equation after multiplication by -1 . The vector $+\nu_j \nabla h_j$ is an algebraic normal term, not the inward physical wall force.

Warning

The sign restriction depends on the convention. If a constraint is rewritten from $h_j \leq 0$ as $-h_j \geq 0$, its normal direction and the corresponding multiplier-sign rule must be changed together.

12.2.2 Complementarity, the Lagrangian, and the KKT first-order conditions

For an inequality wall, the two basic cases are

$$h_j(\mathbf{x}^*) < 0 \implies \nu_j^* = 0, \quad \nu_j^* > 0 \implies h_j(\mathbf{x}^*) = 0.$$

Together with primal and dual feasibility, both are encoded by

$$h_j(\mathbf{x}^*) \leq 0, \quad \nu_j^* \geq 0, \quad \nu_j^* h_j(\mathbf{x}^*) = 0. \quad (12.6)$$

Important note: Complementarity is one-way

An inactive wall cannot exert force, but an active wall need not exert a positive force:

$$h_j(\mathbf{x}^*) = 0 \not\Rightarrow \nu_j^* > 0.$$

The case $c = 1$ in Equation (12.3) is a concrete active but unloaded contact: the particle touches the wall, yet $\nu^* = 0$.

Definition 12.3 (Lagrangian). The Lagrangian of Equation (12.2) is the function

$$L : \mathcal{D} \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}, \quad L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) := f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x}).$$

The Lagrangian is defined for every $\boldsymbol{\nu} \in \mathbb{R}^p$. The restriction $\boldsymbol{\nu} \geq \mathbf{0}$ belongs to the KKT first-order conditions and, later, to dual feasibility; it is not part of the domain of L .

Important note: The Lagrangian is bookkeeping

The identity

$$\nabla_x L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \nabla f(\mathbf{x}) + \sum_i \lambda_i \nabla g_i(\mathbf{x}) + \sum_j \nu_j \nabla h_j(\mathbf{x})$$

packages the force balance and prepares the duality notation. It does not instruct us to minimize L jointly over $(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$.

Definition 12.4 (KKT point and KKT first-order conditions). A feasible point $\mathbf{x}^*$ is a *KKT point* if there are multipliers $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ and $\boldsymbol{\nu}^* \in \mathbb{R}^p$ satisfying

$$\nabla_x L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0}, \quad \text{stationarity,} \quad (12.7)$$

$$g_i(\mathbf{x}^*) = 0, \quad h_j(\mathbf{x}^*) \leq 0, \quad \text{primal feasibility,} \quad (12.8)$$

$$\nu_j^* \geq 0, \quad \text{dual feasibility,} \quad (12.9)$$

$$\nu_j^* h_j(\mathbf{x}^*) = 0, \quad \text{complementary slackness.} \quad (12.10)$$

The four displayed requirements are called the *KKT first-order conditions*, and $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ is called a *KKT triple*.

Warning

Being a KKT point is an algebraic property. The definition contains neither local optimality nor LICQ. A theorem is required to establish these conditions as FONC from local optimality, and additional assumptions are required to deduce optimality from those conditions.

Proposition 12.1 (Active-set form of the KKT first-order conditions). *For a feasible $\mathbf{x}^*$, the KKT first-order conditions are equivalent to the existence of $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ and coefficients $\nu_j^* \geq 0$ for $j \in J(\mathbf{x}^*)$ such that*

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j \in J(\mathbf{x}^*)} \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}. \quad (12.11)$$

Proof. If the full KKT first-order system holds, an inactive index satisfies $h_j(\mathbf{x}^*) < 0$. Complementarity then forces $\nu_j^* = 0$, so stationarity reduces to Equation (12.11). Conversely, extend the active multipliers by setting $\nu_j^* = 0$ for every inactive index. Stationarity and dual feasibility hold by construction. Complementarity holds because active constraints have value zero and inactive constraints have multiplier zero. $\square$

Definition 12.5 (Strict complementarity). A KKT triple $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ satisfies *strict complementarity* (also called strict complementary slackness) if

$$\nu_j^* > 0 \quad \text{for every } j \in J(\mathbf{x}^*).$$

Together with feasibility and complementary slackness, this is equivalent to requiring, for every inequality index j , exactly one of the two cases

$$h_j(\mathbf{x}^*) = 0, \nu_j^* > 0, \quad \text{or} \quad h_j(\mathbf{x}^*) < 0, \nu_j^* = 0.$$

Strict complementarity rules out active but unloaded contact. It is not part of the KKT first-order conditions; it is an additional property of a particular KKT triple and will be useful for the reduced-Hessian discussion in Chapter 13.

12.2.3 Three preliminary KKT first-order calculations

Example 12.2 (A one-dimensional floor). Consider

$$\min_{x \in \mathbb{R}} f(x) = x \quad \text{subject to} \quad h(x) = -x \leq 0.$$

The feasible set is $[0, \infty)$ and its minimizer is $x^* = 0$. Since $f'(0) = 1$ and $h'(0) = -1$, stationarity gives

$$1 - \nu^* = 0,$$

so $\nu^* = 1$. In physical form,

$$\underbrace{-f'(0)}_{-1} + \underbrace{(-\nu^* h'(0))}_{+1} = 0.$$

Example 12.3 (Returning to the opening polygon). At the minimizer of Equation (12.1),

$$J(\mathbf{x}^*) = \{3\}, \quad \nabla f(\mathbf{x}^*) = -\frac{12}{13}(3, 2)^\top, \quad \nabla h_3 = (3, 2)^\top.$$

The active-set stationarity equation is

$$-\frac{12}{13}(3, 2)^\top + \nu_3^*(3, 2)^\top = \mathbf{0},$$

hence

$$\nu_3^* = \frac{12}{13}, \quad \nu_j^* = 0 \quad (j \neq 3).$$

All primal inequalities and multiplier signs hold, and complementarity is immediate.

Example 12.4 (Two active walls). Use the same five polygonal constraints as in Equation (12.1), but minimize

$$f(\mathbf{x}) = (x_1 - 8)^2 + (x_2 - 4)^2.$$

At $\mathbf{x}^* = (4, 3)^\top$, the active set is $J(\mathbf{x}^*) = \{1, 3\}$ and

$$\nabla f(\mathbf{x}^*) = (-8, -2)^\top, \quad \nabla h_1 = (1, 0)^\top, \quad \nabla h_3 = (3, 2)^\top.$$

Stationarity requires

$$(-8, -2)^\top + \nu_1(1, 0)^\top + \nu_3(3, 2)^\top = \mathbf{0}.$$

The second component gives $\nu_3 = 1$, and the first then gives $\nu_1 = 5$. Both multipliers are positive. Thus $-\nabla f(\mathbf{x}^*)$ lies in the cone generated by the active outward normals.

12.3 Two questions and two pathologies

Important note: The two multiplier questions

The force picture leaves two logically distinct questions:

1. Does every constrained local minimizer satisfy the KKT first-order conditions—that is, admit suitable multipliers?
2. If such multipliers exist, are they unique?

Without a regularity assumption, both answers are no.

Example 12.5 (A global minimizer violating the KKT first-order conditions). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) := x_1 + x_2^2 \quad \text{subject to} \quad h(\mathbf{x}) := x_1^2 \leq 0.$$

The constraint forces $x_1 = 0$, so the problem reduces to minimizing x_2^2 . Hence $\mathbf{x}^* = \mathbf{0}$ is the unique global minimizer. However,

$$\nabla f(\mathbf{0}) = (1, 0)^\top, \quad \nabla h(\mathbf{0}) = \mathbf{0}.$$

KKT first-order stationarity would require

$$(1, 0)^\top + \nu \mathbf{0} = \mathbf{0},$$

which is impossible. The constraint has a genuine geometric effect but no first-order normal at the minimizer.

Example 12.6 (Opposing walls and nonunique multipliers). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) := x_1 + x_2^2 \quad \text{subject to} \quad h_1(\mathbf{x}) := x_1 \leq 0, \quad h_2(\mathbf{x}) := -x_1 \leq 0.$$

The two walls force $x_1 = 0$, so $\mathbf{x}^* = \mathbf{0}$ is again the unique global minimizer. Both walls are active. Stationarity is

$$(1, 0)^\top + \nu_1(1, 0)^\top + \nu_2(-1, 0)^\top = \mathbf{0},$$

or $1 + \nu_1 - \nu_2 = 0$. Therefore the complete multiplier set is

$$(\nu_1, \nu_2) = (t, 1 + t), \quad t \geq 0.$$

The active normals are collinear, so the same net reaction has infinitely many nonnegative decompositions.

12.4 LICQ and multiplier uniqueness

Definition 12.6 (Linear independence constraint qualification). LICQ holds at $\mathbf{x} \in \mathcal{S}$ if

$$\{\nabla g_i(\mathbf{x}) : i = 1, \dots, m\} \cup \{\nabla h_j(\mathbf{x}) : j \in J(\mathbf{x})\}$$

is linearly independent.

Equivalently, the matrix having the equality gradients and active inequality gradients as columns has full column rank. Consequently, LICQ is impossible if $m + |J(\mathbf{x})| > n$. Counting is only a preliminary check: even a single active constraint violates LICQ when its gradient vanishes.

Proposition 12.2 (LICQ gives unique multipliers for the KKT first-order conditions). *Suppose $\mathbf{x}^*$ is a KKT point and LICQ holds at $\mathbf{x}^*$. Then its multiplier pair for the KKT first-order conditions is unique.*

Proof. Let $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ and $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\nu}})$ be two multiplier pairs satisfying the KKT first-order conditions. Complementarity sets both inactive multiplier vectors to zero. Subtracting the two stationarity equations yields

$$\sum_{i=1}^m (\lambda_i - \hat{\lambda}_i) \nabla g_i(\mathbf{x}^*) + \sum_{j \in J(\mathbf{x}^*)} (\nu_j - \hat{\nu}_j) \nabla h_j(\mathbf{x}^*) = \mathbf{0}.$$

LICQ makes the displayed active normals linearly independent. Every coefficient is therefore zero. $\square$

Warning

Multiplier uniqueness is not primal-solution uniqueness. LICQ may give a unique force decomposition at each KKT point while the optimization problem still has several distinct minimizers.

Warning

LICQ depends on the displayed constraint functions, not only on the feasible set. For example, $\{0\}$ can be written as $g(x) = x = 0$ or as $\tilde{g}(x) = x^3 = 0$. The first representation satisfies LICQ at zero; the second does not because $\tilde{g}'(0) = 0$. The feasible geometry is identical, but the linearized descriptions differ.

12.5 Tangent and linearized feasible motion

12.5.1 The tangent cone

Definition 12.7 (Tangent cone). The tangent cone to $\mathcal{S}$ at $\mathbf{x} \in \mathcal{S}$ is

$$T_{\mathcal{S}}(\mathbf{x}) := \left\{ \mathbf{d} : \begin{array}{l} \exists t^{(k)} \downarrow 0, \mathbf{x}^{(k)} \in \mathcal{S}, \mathbf{x}^{(k)} \rightarrow \mathbf{x}, \\ (\mathbf{x}^{(k)} - \mathbf{x})/t^{(k)} \rightarrow \mathbf{d} \end{array} \right\}.$$

The sequence definition applies to smooth surfaces, corners, cusps, and isolated points. The numbers $t^{(k)}$ are independent positive scales; they need not equal $\|\mathbf{x}^{(k)} - \mathbf{x}\|$.

Important note: The scale is not a normalization

The same index k labels both the feasible sequence and its positive scale. In general one must not impose $t^{(k)} = \|\mathbf{x}^{(k)} - \mathbf{x}\|$. Doing so would normalize every nonzero limit to unit length and would lose the conic scaling that is essential to $T_{\mathcal{S}}(\mathbf{x})$.

Example 12.7 (Regular and degenerate tangent geometry). At $\mathbf{x}^* = \mathbf{0}$, consider first

$$\mathcal{S}_{\text{reg}} := \{\mathbf{x} : x_1^2 - x_2 \leq 0\} = \{\mathbf{x} : x_2 \geq x_1^2\}.$$

If $\mathbf{d} \in T_{\mathcal{S}_{\text{reg}}}(\mathbf{0})$, every realizing sequence satisfies $x_2^{(k)} \geq (x_1^{(k)})^2 \geq 0$. Division by $t^{(k)} > 0$ and passage to the limit give $d_2 \geq 0$. Conversely, for any $\mathbf{d} = (d_1, d_2)^\top$ with $d_2 \geq 0$, choose $t^{(k)} \downarrow 0$ and set

$$\mathbf{x}^{(k)} := (t^{(k)}d_1, t^{(k)}d_2 + (t^{(k)})^2d_1^2)^\top.$$

Then $(x_1^{(k)})^2 - x_2^{(k)} = -t^{(k)}d_2 \leq 0$ and $\mathbf{x}^{(k)}/t^{(k)} \rightarrow \mathbf{d}$. Hence

$$T_{\mathcal{S}_{\text{reg}}}(\mathbf{0}) = \{\mathbf{d} : d_2 \geq 0\}.$$

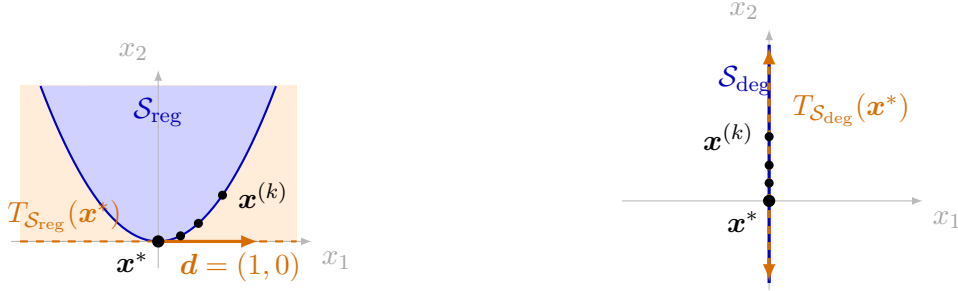
Now consider the degenerate description

$$\mathcal{S}_{\text{deg}} := \{\mathbf{x} : x_1^2 \leq 0\} = \{\mathbf{x} : x_1 = 0\}.$$

Every feasible sequence has first coordinate zero, so every tangent direction has $d_1 = 0$. Conversely, $\mathbf{x}^{(k)} = t^{(k)}(0, d_2)^\top$ realizes every such direction. Therefore

$$T_{\mathcal{S}_{\text{deg}}}(\mathbf{0}) = \{\mathbf{d} : d_1 = 0\}.$$

The two geometries are shown in [Figure 12.2](#). In the regular example the first-order cone is an upper half-plane; in the degenerate example the feasible set and its tangent cone both collapse to the vertical line.



Regular boundary: $\nabla(x_1^2 - x_2)|_0 \neq \mathbf{0}$.

Degenerate boundary: $\nabla(x_1^2)|_0 = \mathbf{0}$.

Figure 12.2: The feasible set (blue) and tangent cone (orange) at the origin for the regular and degenerate examples.

Lemma 12.3 (Elementary properties of the tangent cone). *For every $\mathbf{x} \in \mathcal{S}$, the set $T_{\mathcal{S}}(\mathbf{x})$ contains $\mathbf{0}$, is closed, and is a cone: if $\mathbf{d} \in T_{\mathcal{S}}(\mathbf{x})$ and $\alpha \geq 0$, then $\alpha\mathbf{d} \in T_{\mathcal{S}}(\mathbf{x})$. It need not be convex.*

Proof. The constant sequence proves $\mathbf{0} \in T_{\mathcal{S}}(\mathbf{x})$. If $\alpha > 0$ and $(\mathbf{x}^{(k)} - \mathbf{x})/t^{(k)} \rightarrow \mathbf{d}$, use the same feasible points with $\hat{t}^{(k)} = t^{(k)}/\alpha$ to obtain $(\mathbf{x}^{(k)} - \mathbf{x})/\hat{t}^{(k)} \rightarrow \alpha\mathbf{d}$. The case $\alpha = 0$ follows from the first observation.

For closedness, let $\mathbf{d}^{(k)} \in T_{\mathcal{S}}(\mathbf{x})$ and $\mathbf{d}^{(k)} \rightarrow \mathbf{d}$. For each k , from a sequence realizing $\mathbf{d}^{(k)}$, choose one feasible point $\hat{\mathbf{x}}^{(k)}$ and one scale $\hat{t}^{(k)} \in (0, 1/k)$ such that

$$\|\hat{\mathbf{x}}^{(k)} - \mathbf{x}\| < \frac{1}{k}, \quad \left\| \frac{\hat{\mathbf{x}}^{(k)} - \mathbf{x}}{\hat{t}^{(k)}} - \mathbf{d}^{(k)} \right\| < \frac{1}{k}.$$

After passing to a subsequence for which $\hat{t}^{(k)}$ decreases strictly to zero, these points realize $\mathbf{d}$, proving closedness.

Nonconvexity is possible. For the union of the two coordinate axes in $\mathbb{R}^2$, the tangent cone at the origin is the same union. It contains $(1, 0)^\top$ and $(0, 1)^\top$ but not their midpoint direction. $\square$

Proposition 12.4 (First-order condition on the tangent cone). *If f is differentiable at a local minimizer $\mathbf{x}^*$ of Equation (12.2), then*

$$\nabla f(\mathbf{x}^*)^\top \mathbf{d} \geq 0 \quad \text{for every } \mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*). \quad (12.12)$$

Proof. Let $\mathbf{d} \in T_{\mathcal{S}}(\mathbf{x}^*)$ and choose a defining feasible sequence. For all sufficiently large k , local optimality gives $f(\mathbf{x}^{(k)}) - f(\mathbf{x}^*) \geq 0$. Differentiability gives

$$f(\mathbf{x}^{(k)}) - f(\mathbf{x}^*) = \nabla f(\mathbf{x}^*)^\top (\mathbf{x}^{(k)} - \mathbf{x}^*) + o(\|\mathbf{x}^{(k)} - \mathbf{x}^*\|).$$

Divide by $t^{(k)} > 0$. The quotient $\|\mathbf{x}^{(k)} - \mathbf{x}^*\|/t^{(k)}$ is bounded because $(\mathbf{x}^{(k)} - \mathbf{x}^*)/t^{(k)} \rightarrow \mathbf{d}$. Hence the remainder divided by $t^{(k)}$ tends to zero. Passing to the limit proves Equation (12.12). $\square$

Important note: No Feasible First-Order Descent

Proposition 12.4 is the always-valid first-order necessary condition. It says that the objective has no descent direction among actual feasible velocities. Its difficulty is computational: the tangent cone may be hard to describe directly.

12.5.2 The linearized cone

Definition 12.8 (Linearized feasible-direction cone). At $\mathbf{x} \in \mathcal{S}$, define

$$T_{\text{lin}}(\mathbf{x}) := \left\{ \mathbf{d} : \begin{array}{ll} \nabla g_i(\mathbf{x})^\top \mathbf{d} = 0, & i = 1, \dots, m, \\ \nabla h_j(\mathbf{x})^\top \mathbf{d} \leq 0, & j \in J(\mathbf{x}) \end{array} \right\}. \quad (12.13)$$

The set $T_{\text{lin}}(\mathbf{x})$ is a closed convex polyhedral cone because it is an intersection of finitely many homogeneous linear equalities and inequalities. Unlike $T_{\mathcal{S}}(\mathbf{x})$, it depends on the chosen constraint representation.

Proposition 12.5 (Linearization is an outer approximation). *If the constraints are differentiable at $\mathbf{x} \in \mathcal{S}$, then*

$$T_{\mathcal{S}}(\mathbf{x}) \subseteq T_{\text{lin}}(\mathbf{x}).$$

Proof. Let $\mathbf{d} \in T_{\mathcal{S}}(\mathbf{x})$ and use a defining feasible sequence. For an equality,

$$0 = \frac{g_i(\mathbf{x}^{(k)}) - g_i(\mathbf{x})}{t^{(k)}} = \nabla g_i(\mathbf{x})^\top \frac{\mathbf{x}^{(k)} - \mathbf{x}}{t^{(k)}} + o(1),$$

where the remainder is $o(1)$ because $\|\mathbf{x}^{(k)} - \mathbf{x}\|/t^{(k)}$ is bounded. Hence $\nabla g_i(\mathbf{x})^\top \mathbf{d} = 0$. If $j \in J(\mathbf{x})$, feasibility and $h_j(\mathbf{x}) = 0$ give

$$0 \geq \frac{h_j(\mathbf{x}^{(k)}) - h_j(\mathbf{x})}{t^{(k)}} = \nabla h_j(\mathbf{x})^\top \frac{\mathbf{x}^{(k)} - \mathbf{x}}{t^{(k)}} + \frac{o(\|\mathbf{x}^{(k)} - \mathbf{x}\|)}{t^{(k)}}.$$

The same boundedness argument makes the last term vanish. Taking the limit yields $\nabla h_j(\mathbf{x})^\top \mathbf{d} \leq 0$. Thus $\mathbf{d} \in T_{\text{lin}}(\mathbf{x})$. $\square$

Example 12.8 (The two tangent examples after linearization). For the regular set in [Example 12.7](#),

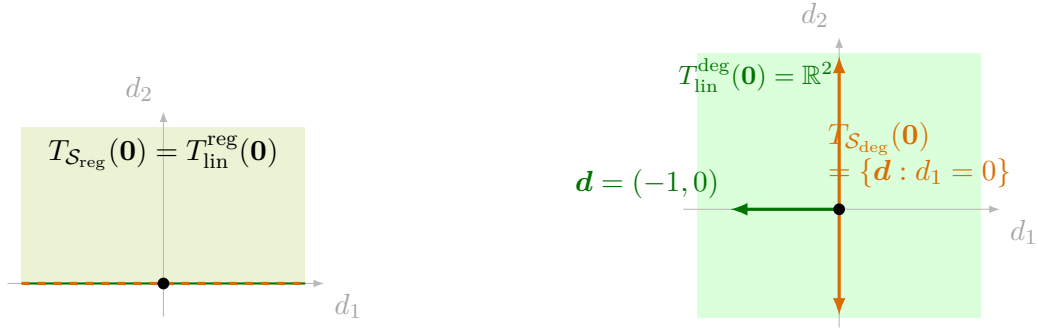
$$\nabla h(\mathbf{0}) = (0, -1)^\top, \quad T_{\text{lin}}^{\text{reg}}(\mathbf{0}) = \{\mathbf{d} : -d_2 \leq 0\} = \{\mathbf{d} : d_2 \geq 0\} = T_{\mathcal{S}_{\text{reg}}}(\mathbf{0}).$$

Thus the constraint gradient predicts exactly the actual first-order motion.

For the degenerate set, $\nabla h(\mathbf{0}) = \mathbf{0}$, and hence

$$T_{\text{lin}}^{\text{deg}}(\mathbf{0}) = \mathbb{R}^2, \quad T_{\mathcal{S}_{\text{deg}}}(\mathbf{0}) = \{\mathbf{d} : d_1 = 0\} \subsetneq T_{\text{lin}}^{\text{deg}}(\mathbf{0}).$$

In particular, $\mathbf{d} = (-1, 0)^\top$ belongs to the linearized cone even though no feasible sequence can realize it. For the objective $f(\mathbf{x}) = x_1 + x_2^2$ from [Example 12.5](#), it is also a fictitious descent direction because $\nabla f(\mathbf{0})^\top \mathbf{d} = -1$.



Regular boundary: both cones are $\{\mathbf{d} : d_2 \geq 0\}$.

Degenerate boundary: the green direction is only linearly feasible.

Figure 12.3: Direction-space comparison: orange is the tangent cone and green is the linearized cone.

Warning

The inclusion $T_{\mathcal{S}}(\mathbf{x}) \subseteq T_{\text{lin}}(\mathbf{x})$ is automatic, but equality is not. At a singular constraint representation, $T_{\text{lin}}(\mathbf{x})$ may contain fictitious directions and must not replace the tangent cone without a valid constraint qualification.

12.5.3 Exact linearization under LICQ

Theorem 12.6 (Exact linearization under LICQ). *Suppose the constraints are continuously differentiable near $\mathbf{x} \in \mathcal{S}$. If LICQ holds at $\mathbf{x}$, then*

$$T_{\mathcal{S}}(\mathbf{x}) = T_{\text{lin}}(\mathbf{x}).$$

Proof. Proposition 12.5 gives one inclusion. For the reverse inclusion, write the active indices as $j_1, \dots, j_s$ and form

$$M = [\nabla g_1(\mathbf{x}) \cdots \nabla g_m(\mathbf{x}) \nabla h_{j_1}(\mathbf{x}) \cdots \nabla h_{j_s}(\mathbf{x})].$$

LICQ says that M has full column rank. Choose a matrix Z whose columns form an orthonormal basis of $\text{Null}(M^\top)$. The $n \times n$ matrix

$$\begin{bmatrix} M^\top \\ Z^\top \end{bmatrix}$$

is nonsingular: a vector in its null space is orthogonal to both $\text{Range}(M)$ and $\text{Null}(M^\top)$, two orthogonal complementary subspaces.

Fix $\mathbf{d} \in T_{\text{lin}}(\mathbf{x})$ and define, for $\mathbf{z}$ near $\mathbf{x}$,

$$\Phi(\mathbf{z}) := (g_1(\mathbf{z}), \dots, g_m(\mathbf{z}), h_{j_1}(\mathbf{z}), \dots, h_{j_s}(\mathbf{z}), Z^\top(\mathbf{z} - \mathbf{x})).$$

Its derivative at $\mathbf{x}$ is the nonsingular matrix just displayed. By the inverse function theorem, Φ maps a neighborhood of $\mathbf{x}$ diffeomorphically onto a neighborhood of $\Phi(\mathbf{x})$. Hence, for every sufficiently small t , there is a unique nearby $\mathbf{z}(t)$ satisfying

$$\Phi(\mathbf{z}(t)) = (0, \dots, 0, t\nabla h_{j_1}(\mathbf{x})^\top \mathbf{d}, \dots, t\nabla h_{j_s}(\mathbf{x})^\top \mathbf{d}, tZ^\top \mathbf{d}).$$

Because $\mathbf{d} \in T_{\text{lin}}(\mathbf{x})$, every prescribed active inequality value is nonpositive for $t > 0$. Inactive inequalities remain strict by continuity. Therefore $\mathbf{z}(t) \in \mathcal{S}$ for all sufficiently small $t > 0$.

Differentiating the identity for $\Phi(\mathbf{z}(t))$ at $t = 0$ shows

$$M^\top \mathbf{z}'(0) = M^\top \mathbf{d}, \quad Z^\top \mathbf{z}'(0) = Z^\top \mathbf{d}.$$

Nonsingularity implies $\mathbf{z}'(0) = \mathbf{d}$. Thus $(\mathbf{z}(t) - \mathbf{x})/t \rightarrow \mathbf{d}$, so $\mathbf{d} \in T_{\mathcal{S}}(\mathbf{x})$. $\square$

Important note: The Two Roles of LICQ

LICQ performs two different jobs:

1. *geometric regularity*: it gives $T_{\mathcal{S}}(\mathbf{x}) = T_{\text{lin}}(\mathbf{x})$, so the gradient model describes all actual first-order restrictions;
2. *algebraic identifiability*: if KKT multipliers exist for the KKT first-order conditions, linear independence makes their normal-force representation unique.

Neither role says that the primal minimizer itself is unique.

Warning

LICQ is sufficient, not necessary, for exact linearization. The duplicated description $-x \leq 0, -2x \leq 0$ fails LICQ at zero, but both the tangent and linearized cones equal $[0, \infty)$. What fails there is uniqueness of the normal-force coefficients, not first-order geometry.

12.6 From feasible directions to the KKT FONC

12.6.1 Polar cones and a Farkas alternative

Definition 12.9 (Polar and regular normal cones). For a cone $K \subseteq \mathbb{R}^n$, its negative polar is

$$K^\circ := \{\mathbf{v} : \mathbf{v}^\top \mathbf{d} \leq 0 \text{ for every } \mathbf{d} \in K\}.$$

The regular normal cone to $\mathcal{S}$ at $\mathbf{x}$ is

$$N_{\mathcal{S}}(\mathbf{x}) := T_{\mathcal{S}}(\mathbf{x})^\circ.$$

The tangent-cone condition Equation (12.12) is equivalently

$$-\nabla f(\mathbf{x}^*) \in N_{\mathcal{S}}(\mathbf{x}^*).$$

Lemma 12.7 (A finitely generated cone is closed). If $\mathbf{a}_1, \dots, \mathbf{a}_q \in \mathbb{R}^n$, then

$$K := \left\{ \sum_{r=1}^q \alpha_r \mathbf{a}_r : \alpha_r \geq 0 \right\}$$

is closed.

Proof. Every point of K has a representation using linearly independent generators. Indeed, suppose $\sum_{r \in I} \alpha_r \mathbf{a}_r$ uses a linearly dependent subfamily with every $\alpha_r > 0$. Choose a nonzero vector γ such that $\sum_{r \in I} \gamma_r \mathbf{a}_r = \mathbf{0}$, and reverse its sign if necessary so that some $\gamma_r > 0$. With

$$t := \min_{\{r \in I: \gamma_r > 0\}} \frac{\alpha_r}{\gamma_r},$$

the coefficients $\alpha_r - t\gamma_r$ are all nonnegative and at least one is zero, while the represented vector is unchanged. Repeating this reduction leaves at most n linearly independent generators.

Now let $\mathbf{x}^{(k)} \in K$ and $\mathbf{x}^{(k)} \rightarrow \mathbf{x}$. There are only finitely many subsets of the generators, so a subsequence uses one fixed linearly independent subfamily. If that subfamily is empty, the subsequence is identically zero and $\mathbf{x} = \mathbf{0} \in K$. Otherwise, let its vectors be the columns of A and write $\mathbf{x}^{(k)} = A\boldsymbol{\alpha}^{(k)}$ with $\boldsymbol{\alpha}^{(k)} \geq \mathbf{0}$. Because A has full column rank,

$$\boldsymbol{\alpha}^{(k)} = (A^\top A)^{-1} A^\top \mathbf{x}^{(k)}$$

converges to a nonnegative vector $\boldsymbol{\alpha}$. Hence $\mathbf{x} = A\boldsymbol{\alpha} \in K$. $\square$

Lemma 12.8 (Cone alternative: a Farkas form). *Let $H \in \mathbb{R}^{n \times s}$, $G \in \mathbb{R}^{n \times m}$, and $\mathbf{u} \in \mathbb{R}^n$. Exactly one of the following alternatives holds:*

1. $\mathbf{u} = H\mathbf{y} + G\mathbf{v}$ for some $\mathbf{y} \leq \mathbf{0}$ and $\mathbf{v} \in \mathbb{R}^m$;
2. there is $\mathbf{d}$ such that $\mathbf{u}^\top \mathbf{d} < 0$, $H^\top \mathbf{d} \leq \mathbf{0}$, and $G^\top \mathbf{d} = \mathbf{0}$.

Proof. Let

$$K := \{H\mathbf{y} + G\mathbf{v} : \mathbf{y} \leq \mathbf{0}, \mathbf{v} \in \mathbb{R}^m\}.$$

If the columns of H are $\mathbf{h}_r$ and those of G are $\mathbf{g}_i$, then

$$K = \text{cone}\{-\mathbf{h}_1, \dots, -\mathbf{h}_s, \mathbf{g}_1, -\mathbf{g}_1, \dots, \mathbf{g}_m, -\mathbf{g}_m\}.$$

Thus K is closed by Lemma 12.7; it is also convex.

If alternative 1 holds, every $\mathbf{d}$ satisfying the last two relations in alternative 2 obeys

$$\mathbf{u}^\top \mathbf{d} = \mathbf{y}^\top H^\top \mathbf{d} + \mathbf{v}^\top G^\top \mathbf{d} \geq 0,$$

so the alternatives are mutually exclusive.

Suppose alternative 1 fails. Let $\bar{\mathbf{u}}$ be the Euclidean projection of $\mathbf{u}$ onto K , and set $\mathbf{d} := \bar{\mathbf{u}} - \mathbf{u}$. The projection inequality gives

$$(\mathbf{u} - \bar{\mathbf{u}})^\top (\mathbf{z} - \bar{\mathbf{u}}) \leq 0 \quad (\mathbf{z} \in K).$$

Using $\mathbf{z} = \mathbf{0}$ and $\mathbf{z} = 2\bar{\mathbf{u}}$ shows $(\mathbf{u} - \bar{\mathbf{u}})^\top \bar{\mathbf{u}} = 0$. Hence $\mathbf{d}^\top \mathbf{z} \geq 0$ for every $\mathbf{z} \in K$. Testing the generators $-H\mathbf{e}_r$, $G\mathbf{e}_i$, and $-G\mathbf{e}_i$ gives

$$H^\top \mathbf{d} \leq \mathbf{0}, \quad G^\top \mathbf{d} = \mathbf{0}.$$

Finally,

$$\mathbf{u}^\top \mathbf{d} = -\|\mathbf{u} - \bar{\mathbf{u}}\|^2 < 0$$

because $\mathbf{u} \notin K$. Alternative 2 therefore holds. $\square$

Proposition 12.9 (Polar of the linearized cone). *Let G have columns $\nabla g_i(\mathbf{x})$, and let H have columns $\nabla h_j(\mathbf{x})$ for $j \in J(\mathbf{x})$. Then*

$$T_{\text{lin}}(\mathbf{x})^\circ = \{G\boldsymbol{\lambda} + H\boldsymbol{\nu} : \boldsymbol{\lambda} \in \mathbb{R}^m, \boldsymbol{\nu} \geq \mathbf{0}\}. \quad (12.14)$$

Proof. If $\mathbf{v} = G\boldsymbol{\lambda} + H\boldsymbol{\nu}$ with $\boldsymbol{\nu} \geq \mathbf{0}$ and $\mathbf{d} \in T_{\text{lin}}(\mathbf{x})$, then

$$\mathbf{v}^\top \mathbf{d} = \boldsymbol{\lambda}^\top G^\top \mathbf{d} + \boldsymbol{\nu}^\top H^\top \mathbf{d} \leq 0.$$

Thus the right-hand side of Equation (12.14) is contained in the left.

Conversely, suppose $\mathbf{v} \in T_{\text{lin}}(\mathbf{x})^\circ$ but has no representation on the right. Then $-\mathbf{v}$ has no representation $H\mathbf{y} + G\boldsymbol{\lambda}$ with $\mathbf{y} \leq \mathbf{0}$. Apply Lemma 12.8 with $\mathbf{u} = -\mathbf{v}$. It gives $\mathbf{d}$ satisfying

$$-\mathbf{v}^\top \mathbf{d} < 0, \quad H^\top \mathbf{d} \leq \mathbf{0}, \quad G^\top \mathbf{d} = \mathbf{0}.$$

Thus $\mathbf{d} \in T_{\text{lin}}(\mathbf{x})$ but $\mathbf{v}^\top \mathbf{d} > 0$, contradicting the definition of the polar. $\square$

12.6.2 KKT FONC under LICQ

Theorem 12.10 (KKT first-order necessary conditions (FONC)). *Let $\mathbf{x}^*$ be a local minimizer of Equation (12.2). If LICQ holds at $\mathbf{x}^*$, then $\mathbf{x}^*$ is a KKT point, and its KKT multipliers are unique.*

Proof. Proposition 12.4 gives

$$-\nabla f(\mathbf{x}^*) \in T_S(\mathbf{x}^*)^\circ.$$

By Theorem 12.6, $T_S(\mathbf{x}^*) = T_{\text{lin}}(\mathbf{x}^*)$. Therefore

$$-\nabla f(\mathbf{x}^*) \in T_{\text{lin}}(\mathbf{x}^*)^\circ.$$

Proposition 12.9 supplies $\boldsymbol{\lambda}^*$ and nonnegative active multipliers such that

$$-\nabla f(\mathbf{x}^*) = \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j \in J(\mathbf{x}^*)} \nu_j^* \nabla h_j(\mathbf{x}^*).$$

Set inactive multipliers to zero. This gives stationarity, dual feasibility, and complementarity; primal feasibility is already known. Hence $\mathbf{x}^*$ is a KKT point. Uniqueness follows from Proposition 12.2. $\square$

Important note: The Exact Logical Implication

$$\begin{array}{c} \text{local minimizer + LICQ} \\ \implies \text{KKT FONC} \\ \text{with unique multipliers} \end{array}$$

Local optimality gives no descent on the true tangent cone; LICQ replaces that cone by its linear model; Farkas converts the resulting directional inequality into an algebraic force balance.

Corollary 12.11 (Unconstrained, interior, and equality-only cases). *Under the hypotheses of Theorem 12.10:*

1. if $m = p = 0$, then $\nabla f(\mathbf{x}^*) = \mathbf{0}$;
2. if there are no equalities and $J(\mathbf{x}^*) = \emptyset$, then $\nabla f(\mathbf{x}^*) = \mathbf{0}$;
3. if only equalities are present, then $-\nabla f(\mathbf{x}^*)$ belongs to $\text{span}\{\nabla g_i(\mathbf{x}^*)\}_{i=1}^m$;
4. with one active inequality and no equalities, $-\nabla f(\mathbf{x}^*) = \nu^* \nabla h(\mathbf{x}^*)$ for a unique $\nu^* \geq 0$.

Proof. Each statement follows by deleting the absent terms from Equation (12.11). In the unconstrained or inequality-interior cases, $T_{\text{lin}}(\mathbf{x}^*) = \mathbb{R}^n$, so the tangent-cone condition also directly forces $\nabla f(\mathbf{x}^*) = \mathbf{0}$. LICQ gives uniqueness of the listed coefficients in cases 3 and 4. $\square$

12.7 KKT first-order calculations and logical traps

12.7.1 A practical active-set workflow

For a small problem, the following procedure uses the KKT first-order conditions as a candidate filter and converts the system into manageable cases.

1. If $\mathbf{x}$ is given, compute its active set. If $\mathbf{x}$ is unknown, guess an active set $\hat{J} \subseteq \{1, \dots, p\}$.
2. Solve stationarity, all equality constraints, and $h_j(\mathbf{x}) = 0$ for $j \in \hat{J}$.
3. Set $\nu_j = 0$ for every $j \notin \hat{J}$.
4. Check every omitted inequality, all multiplier signs, and every complementarity equation.
5. Retain the surviving points satisfying the KKT first-order conditions and use further information to classify them.

The fourth step is essential: the equations in the second step do not remember the one-sided nature of an inequality. A solution with a negative inequality multiplier or a violated inactive constraint must be discarded.

Example 12.9 (A complete KKT check for a candidate). Consider

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & -3x_1 - 2x_2 \\ \text{subject to} \quad & h_1(\mathbf{x}) := x_1 + x_2 - 1 \leq 0, \\ & h_2(\mathbf{x}) := -x_1 \leq 0, \quad h_3(\mathbf{x}) := -x_2 \leq 0. \end{aligned} \tag{12.15}$$

The objective improves in the direction $(3, 2)^\top$, which suggests the vertex $\mathbf{x}^* = (1, 0)^\top$. Its active set is $J(\mathbf{x}^*) = \{1, 3\}$. The active gradients are

$$\nabla h_1 = (1, 1)^\top, \quad \nabla h_3 = (0, -1)^\top,$$

so stationarity becomes

$$(-3, -2)^\top + \nu_1(1, 1)^\top + \nu_3(0, -1)^\top = \mathbf{0}.$$

It follows that

$$\nu_1^* = 3, \quad \nu_3^* = 1, \quad \nu_2^* = 0.$$

We now perform all checks. Primal feasibility follows from

$$h_1(\mathbf{x}^*) = 0, \quad h_2(\mathbf{x}^*) = -1 < 0, \quad h_3(\mathbf{x}^*) = 0.$$

Every multiplier is nonnegative, and

$$\nu_1^* h_1(\mathbf{x}^*) = \nu_2^* h_2(\mathbf{x}^*) = \nu_3^* h_3(\mathbf{x}^*) = 0.$$

Thus $\mathbf{x}^*$ is a KKT point. In this particular example, global optimality can also be verified directly: feasibility gives $x_1, x_2 \geq 0$ and $x_1 + x_2 \leq 1$, hence

$$3x_1 + 2x_2 \leq 3(x_1 + x_2) \leq 3.$$

Therefore $-3x_1 - 2x_2 \geq -3 = f(\mathbf{x}^*)$, with equality only at $\mathbf{x}^* = (1, 0)^\top$.

Important note: KKT first-order conditions are filters, not certificates

An active-set calculation can locate points satisfying the first-order balance law. It does not, in a general nonconvex problem, classify those points. Direct comparison, second-order conditions, convexity, or some other argument is still needed to prove local or global minimality.

Warning

Active-set enumeration is mainly a reasoning tool for modest problems. With p inequalities there are as many as 2^p candidate active sets, and each resulting nonlinear system may have several solutions or no closed-form solution. Numerical constrained algorithms therefore do not usually enumerate every subset blindly.

12.7.2 Two logical traps

Example 12.10 (KKT FONC and LICQ do not imply a minimum). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) := -x_2 \quad \text{subject to} \quad h(\mathbf{x}) := x_2 - x_1^2 \leq 0.$$

At $\mathbf{x}^* = \mathbf{0}$,

$$\nabla f(\mathbf{x}^*) = (0, -1)^\top, \quad \nabla h(\mathbf{x}^*) = (0, 1)^\top.$$

The multiplier $\nu^* = 1$ satisfies stationarity, dual feasibility, and complementarity. LICQ also holds because the single active gradient is nonzero. Nevertheless, the feasible curve $\mathbf{x}(t) = (t, t^2)^\top$ satisfies

$$f(\mathbf{x}(t)) = -t^2 < 0 = f(\mathbf{x}^*) \quad (t \neq 0).$$

Thus the origin is not a local minimizer. The KKT FONC are necessary under LICQ, but they are not sufficient in a general nonconvex problem.

Example 12.11 (LICQ is sufficient, not necessary). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} -x_1 + x_2^2 \quad \text{subject to} \quad h_1(\mathbf{x}) := x_1 \leq 0, \quad h_2(\mathbf{x}) := 2x_1 \leq 0.$$

For every feasible point, $-x_1 + x_2^2 \geq 0$, with equality only at the origin. Hence $\mathbf{x}^* = \mathbf{0}$ is the unique global minimizer. Both constraints are active, but

$$\nabla h_2(\mathbf{0}) = 2\nabla h_1(\mathbf{0}),$$

so LICQ fails. The KKT first-order conditions nevertheless hold for every pair $(\nu_1, \nu_2) \geq \mathbf{0}$ satisfying

$$-1 + \nu_1 + 2\nu_2 = 0.$$

For example, $(\nu_1, \nu_2) = (1, 0)$ and $(0, 1/2)$ are two distinct multiplier pairs. Failure of LICQ therefore implies neither failure of the KKT first-order conditions nor failure of optimality; it merely removes the guarantees supplied by [Theorem 12.10](#) and [proposition 12.2](#).

Warning

The most common first-order mistakes are:

1. changing $h_j \leq 0$ to $h_j \geq 0$ without changing the multiplier-sign rule;
2. calling $+\nu_j \nabla h_j$ the physical wall force, although with our convention the physical force is $-\nu_j \nabla h_j$;
3. assuming that an active constraint must have a positive multiplier;
4. treating every KKT point as a minimizer;
5. treating LICQ as necessary, or checking linear independence for inactive inequalities as well as active ones;
6. replacing $T_{\mathcal{S}}$ by T_{lin} at a singular point without a valid constraint qualification.

Multipliers also depend on how a constraint is scaled. If $\tilde{h}_j = a_j h_j$ with $a_j > 0$, then the same feasible set is obtained, and the corresponding multiplier is $\tilde{\nu}_j = \nu_j / a_j$. The normal contribution is invariant:

$$\tilde{\nu}_j \nabla \tilde{h}_j = \frac{\nu_j}{a_j} a_j \nabla h_j = \nu_j \nabla h_j.$$

Important note: What Is Geometric and What Is Representational

The feasible set, tangent cone, and total normal force are geometric objects. The linearized cone, LICQ status, and individual multiplier values can depend on the chosen constraint representation. Conclusions involving the latter must therefore be read together with their hypotheses.

12.8 Optional supplement: Fritz–John multipliers

The preceding LICQ–tangent-cone–Farkas argument is sufficient for the rest of the course. For completeness, this optional supplement records the more general Fritz–John rule. It

explains why an objective coefficient may be zero before a constraint qualification is imposed and why LICQ permits its normalization to one. The result is not needed in order to use the KKT first-order conditions developed above.

Lemma 12.12 (Gordan alternative relative to equalities). *Let $q \geq 1$, let $B \in \mathbb{R}^{n \times q}$, and let $C \in \mathbb{R}^{n \times m}$ have full column rank. Exactly one of the following holds:*

1. *there are $\mathbf{y} \geq \mathbf{0}$, $\mathbf{y} \neq \mathbf{0}$, and $\boldsymbol{\lambda} \in \mathbb{R}^m$ such that $B\mathbf{y} + C\boldsymbol{\lambda} = \mathbf{0}$;*
2. *there is $\mathbf{d} \in \mathbb{R}^n$ such that $B^\top \mathbf{d} < \mathbf{0}$ componentwise and $C^\top \mathbf{d} = \mathbf{0}$.*

Proof. Let P be the orthogonal projector onto $\text{Null}(C^\top)$, and let $\mathbf{b}_1, \dots, \mathbf{b}_q$ be the columns of B . Consider the compact convex set

$$Q := \text{conv}\{P\mathbf{b}_1, \dots, P\mathbf{b}_q\}.$$

If $\mathbf{0} \in Q$, then some $\mathbf{y} \geq \mathbf{0}$ with $\mathbf{1}^\top \mathbf{y} = 1$ satisfies $P B \mathbf{y} = \mathbf{0}$. Hence $B \mathbf{y} \in \text{Null}(P) = \text{Range}(C)$, so a $\boldsymbol{\lambda}$ exists with $B \mathbf{y} + C \boldsymbol{\lambda} = \mathbf{0}$. This is alternative 1.

If $\mathbf{0} \notin Q$, let $\mathbf{z}$ be the projection of $\mathbf{0}$ onto Q . Then $\mathbf{z} \neq \mathbf{0}$, and the projection inequality gives

$$\mathbf{z}^\top \mathbf{w} \geq \|\mathbf{z}\|^2 \quad (\mathbf{w} \in Q).$$

Set $\mathbf{d} = -\mathbf{z}$. Since $\mathbf{z} \in \text{Null}(C^\top)$, $C^\top \mathbf{d} = \mathbf{0}$. Also, for each r ,

$$\mathbf{b}_r^\top \mathbf{d} = (P \mathbf{b}_r)^\top (-\mathbf{z}) \leq -\|\mathbf{z}\|^2 < 0.$$

Thus alternative 2 holds.

Finally, the alternatives cannot hold simultaneously: if they did, then

$$0 = (B \mathbf{y} + C \boldsymbol{\lambda})^\top \mathbf{d} = \mathbf{y}^\top B^\top \mathbf{d} < 0,$$

because $\mathbf{y} \geq \mathbf{0}$, $\mathbf{y} \neq \mathbf{0}$, and every component of $B^\top \mathbf{d}$ is negative. □

Theorem 12.13 (Fritz–John necessary conditions). *Let $\mathbf{x}^*$ be a local minimizer of Equation (12.2). Then there exist*

$$\mu_0 \geq 0, \quad \boldsymbol{\mu} \in \mathbb{R}^m, \quad \boldsymbol{\eta} \in \mathbb{R}_+^p,$$

not all zero, such that

$$\mu_0 \nabla f(\mathbf{x}^*) + \sum_{i=1}^m \mu_i \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \eta_j \nabla h_j(\mathbf{x}^*) = \mathbf{0}, \quad (12.16)$$

$$\eta_j h_j(\mathbf{x}^*) = 0, \quad j = 1, \dots, p. \quad (12.17)$$

Proof. Let C collect the equality gradients at $\mathbf{x}^*$. If these columns are linearly dependent, choose $\boldsymbol{\mu} \neq \mathbf{0}$ with $C \boldsymbol{\mu} = \mathbf{0}$, and set $\mu_0 = 0$ and $\boldsymbol{\eta} = \mathbf{0}$. Then Equations (12.16) and (12.17) hold.

Now suppose C has full column rank. Let B have first column $\nabla f(\mathbf{x}^*)$ and remaining columns $\nabla h_j(\mathbf{x}^*)$ for $j \in J(\mathbf{x}^*)$. Apply Lemma 12.12. If its first alternative holds, write $\mathbf{y} = (\mu_0, \boldsymbol{\eta}_J)$, and extend $\boldsymbol{\eta}_J$ by zeros on inactive indices. This gives all the required relations and nontriviality.

It remains to rule out the second alternative. If it held, there would be a direction $\mathbf{d}$ such that

$$\nabla f(\mathbf{x}^*)^\top \mathbf{d} < 0, \quad C^\top \mathbf{d} = \mathbf{0}, \quad \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} < 0 \quad (j \in J(\mathbf{x}^*)).$$

Because C has full column rank, choose a matrix Z whose columns form an orthonormal basis of $\text{Null}(C^\top)$. Define

$$\Phi(\mathbf{z}) := (g_1(\mathbf{z}), \dots, g_m(\mathbf{z}), Z^\top(\mathbf{z} - \mathbf{x}^*)).$$

The derivative $D\Phi(\mathbf{x}^*) = [C^\top; Z^\top]$ is nonsingular, so the inverse function theorem gives a local C^1 inverse. Since $C^\top \mathbf{d} = \mathbf{0}$, for sufficiently small t define

$$\mathbf{x}(t) := \Phi^{-1}(0, \dots, 0, tZ^\top \mathbf{d}).$$

Then $g_i(\mathbf{x}(t)) = 0$, $\mathbf{x}(0) = \mathbf{x}^*$, and differentiation at zero gives

$$C^\top \mathbf{x}'(0) = \mathbf{0} = C^\top \mathbf{d}, \quad Z^\top \mathbf{x}'(0) = Z^\top \mathbf{d}.$$

The nonsingularity of $[C^\top; Z^\top]$ therefore implies $\mathbf{x}'(0) = \mathbf{d}$. For every active inequality,

$$h_j(\mathbf{x}(t)) = t \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} + o(t) < 0$$

for all sufficiently small $t > 0$; inactive inequalities remain strict by continuity. Thus $\mathbf{x}(t)$ is feasible. At the same time,

$$f(\mathbf{x}(t)) = f(\mathbf{x}^*) + t \nabla f(\mathbf{x}^*)^\top \mathbf{d} + o(t) < f(\mathbf{x}^*)$$

for small $t > 0$, contradicting local minimality. Therefore the first alternative must hold. $\square$

Corollary 12.14 (LICQ excludes abnormality). *If LICQ holds at the local minimizer $\mathbf{x}^*$, then every nontrivial Fritz–John multiplier system has $\mu_0 > 0$. Dividing Equation (12.16) by μ_0 yields the KKT FONC.*

Proof. Suppose $\mu_0 = 0$. Complementarity removes inactive terms, so Equation (12.16) becomes

$$\sum_i \mu_i \nabla g_i(\mathbf{x}^*) + \sum_{j \in J(\mathbf{x}^*)} \eta_j \nabla h_j(\mathbf{x}^*) = \mathbf{0}.$$

LICQ forces every μ_i and active η_j to be zero. Inactive multipliers are already zero, contradicting nontriviality. Hence $\mu_0 > 0$. Set $\lambda_i = \mu_i/\mu_0$ and $\nu_j = \eta_j/\mu_0$. $\square$

Important note: Optional perspective on normalization

Fritz–John is the unnormalized multiplier rule. LICQ eliminates the abnormal case $\mu_0 = 0$, after which the objective coefficient can be scaled to one. Thus the usual KKT normalization is a consequence of regularity rather than an additional optimality assumption.

Chapter summary

- Constraints replace the unconstrained equation $\nabla f = \mathbf{0}$ by a balance among the objective force, equality reactions, and active inequality reactions.
- For $h_j \leq 0$, dual feasibility $\nu_j \geq 0$ expresses one-sided reaction, while $\nu_j h_j = 0$ expresses the contact logic. An active constraint may still have zero multiplier.
- The tangent-cone condition is always necessary at a local minimizer. The linearized cone is a computable outer approximation, and LICQ makes the two cones equal.
- Farkas' alternative converts the absence of feasible first-order descent into the stationarity part of the KKT FONC. Hence

local minimizer + LICQ $\implies$ KKT FONC with unique multipliers.

- The KKT FONC are generally necessary rather than sufficient, and LICQ is sufficient rather than necessary for their guarantee. The next chapter asks which feasible curvatures make a balanced point a strict local minimum.

12.9 Exercises

Exercise 12.1 (KKT first-order signs and constraint rescaling). Consider

$$\min_{x \in \mathbb{R}} \frac{1}{2}(x-2)^2 \quad \text{subject to} \quad h(x) := x \leq 0.$$

1. Find the global minimizer directly.
2. Write the complete KKT first-order system and determine the multiplier.
3. Replace the constraint by the equivalent constraint $\hat{h}(x) := 4x \leq 0$. Determine the new multiplier and verify that $\nu h'(x^*) = \hat{\nu} \hat{h}'(x^*)$.

Exercise 12.2 (Active sets and linearized cones). Let

$$\mathcal{S} := \{\mathbf{x} \in \mathbb{R}^2 : h_1(\mathbf{x}) := x_1^2 + x_2^2 - 4 \leq 0, \quad h_2(\mathbf{x}) := -x_2 \leq 0\}.$$

At each of

$$\mathbf{x}^a = (2, 0)^\top, \quad \mathbf{x}^b = (0, 2)^\top,$$

determine the active set, decide whether LICQ holds, and compute $T_{\text{lin}}(\mathbf{x}^a)$ and $T_{\text{lin}}(\mathbf{x}^b)$. Sketch each cone. Finally, use a theorem from this chapter to determine the corresponding tangent cones.

Exercise 12.3 (An equality and an active inequality). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} \frac{1}{2}((x_1+1)^2 + (x_2-2)^2) \quad \text{subject to} \quad g(\mathbf{x}) := x_1 + x_2 - 1 = 0, \quad h(\mathbf{x}) := -x_1 \leq 0.$$

1. Eliminate the equality constraint and find the minimizer directly.

2. Determine the active set and verify LICQ at the minimizer.
3. Compute the equality and inequality multipliers and verify every part of the KKT system.
4. Interpret the stationarity equation as a force balance.

Exercise 12.4 (Redundant equalities). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} (x_1 - 1)^2 + (x_2 - 1)^2 \quad \text{subject to} \quad g_1(\mathbf{x}) := x_1 + x_2 = 0, \quad g_2(\mathbf{x}) := 2x_1 + 2x_2 = 0.$$

1. Find the unique global minimizer by restricting the objective to the feasible line.
2. Show that LICQ fails at every feasible point.
3. Compute the complete set of KKT multiplier pairs at the minimizer.
4. Compute the tangent and linearized cones at the minimizer and show that they are nevertheless equal.
5. Explain why these conclusions contradict none of the theorems in this chapter.

Exercise 12.5 (Complete active-set enumeration). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) := (x_1 - 4)^2 + (x_2 - 1)^2$$

subject to

$$h_1(\mathbf{x}) := x_1 + 2x_2 - 2 \leq 0, \quad h_2(\mathbf{x}) := -x_1 \leq 0, \quad h_3(\mathbf{x}) := -x_2 \leq 0.$$

1. Enumerate the feasible active-set possibilities and find every KKT point, discarding any solution that violates primal feasibility, dual feasibility, or complementary slackness.
2. Show that the only KKT point is $\mathbf{x}^* = (2, 0)^\top$ and determine all three multipliers.
3. Without using a sufficiency theorem based on the KKT first-order conditions, verify directly that $\mathbf{x}^*$ is the unique global minimizer by using the exact quadratic expansion of f about $\mathbf{x}^*$.

Exercise 12.6 (An active but unloaded constraint). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} -x_1 \quad \text{subject to} \quad g(\mathbf{x}) := x_1^2 + x_2^2 - 1 = 0, \quad h(\mathbf{x}) := x_2 \leq 0.$$

1. Show directly that $\mathbf{x}^* = (1, 0)^\top$ is the unique global minimizer.
2. Verify LICQ and compute $T_{\text{lin}}(\mathbf{x}^*)$.
3. For every $\mathbf{d} = (0, d_2)^\top$ with $d_2 \leq 0$, construct an explicit feasible curve through $\mathbf{x}^*$ whose derivative at zero is $\mathbf{d}$. Deduce directly that $T_S(\mathbf{x}^*) = T_{\text{lin}}(\mathbf{x}^*)$.
4. Compute the KKT multipliers and explain why the active inequality has zero multiplier.

Exercise 12.7 (A polar cone and a Farkas certificate). Let

$$K := \{\mathbf{d} \in \mathbb{R}^3 : d_1 + d_3 = 0, d_2 \leq 0, d_3 \leq 0\}.$$

1. Use [Proposition 12.9](#) to prove that

$$K^\circ = \left\{ \lambda(1, 0, 1)^\top + \nu_1(0, 1, 0)^\top + \nu_2(0, 0, 1)^\top : \lambda \in \mathbb{R}, \nu_1, \nu_2 \geq 0 \right\}.$$

2. Show that $\mathbf{u} = (2, 1, 3)^\top$ belongs to K° by finding its multiplier representation.
3. Show that $\mathbf{v} = (2, -1, 3)^\top$ does not belong to K° . Find $\mathbf{d} \in K$ such that $\mathbf{v}^\top \mathbf{d} > 0$, and explain why this direction is a Farkas separation certificate.

Exercise 12.8 (Supplementary: an abnormal Fritz–John point). Consider

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) := x_2 \quad \text{subject to} \quad h(\mathbf{x}) := x_1^2 - x_2^3 \leq 0.$$

1. Prove that $\mathbf{x}^* = \mathbf{0}$ is the unique global minimizer.
2. Prove directly from the sequence definition that

$$T_S(\mathbf{0}) = \{(0, d_2)^\top : d_2 \geq 0\}.$$

3. Compute $T_{\text{lin}}(\mathbf{0})$ and explain the strict inclusion $T_S(\mathbf{0}) \subsetneq T_{\text{lin}}(\mathbf{0})$.
4. Verify the tangent-cone first-order necessary condition.
5. Show that no KKT multiplier exists.
6. Find a nontrivial Fritz–John multiplier system and prove that every Fritz–John system at this point is abnormal.
7. Identify precisely where LICQ fails and explain why the objective coefficient cannot be normalized to one.